

C. SAI HALLIE

Software Engineer | Audio Systems & Backend Tooling

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PROFILE

Software engineer with a systems-first approach, working across real-time audio, backend data pipelines, and full-stack web. Comfortable in C++, C#, and Python, and experienced building tooling where correctness, testability, and clear failure behaviour matters more than speed of delivery. Currently based in Japan and seeking a full-time engineering role.

TECHNICAL SKILLS

Languages C++, C#, Python, SQL, JavaScript, HTML/CSS

Audio Engineering Wwise, FMOD, Unreal Engine 4/5, Ableton Live, Pro Tools

Data & Backend SQLite, schema design (3NF), ETL pipelines, REST API integration

Web & Tooling Git, GitHub Pages, Cloudflare CDN, sql.js / WebAssembly, DNS configuration

Languages Spoken English (native), Japanese (JPT 365, conversational; actively studying)

EXPERIENCE

Apple Cider Studios [Dec 2022] – Feb 2025

[Lead Audio Engineer] · Part-time

- Implement and debug real-time audio systems, working across engine and middleware integration layers.
- Diagnose reproduction-resistant issues in interactive systems, tracing faults across component boundaries.
- Build supporting tooling and scripts to speed up iteration and validation.

Independent Music Production & Sound Design 2020 – Present

Freelance

- Deliver original music and sound design for independent clients under a self-managed production pipeline.
- Manage project scoping, revision cycles, and delivery independently.

SELECTED PROJECTS

Spatial Audio Demo — Unreal Engine 5 + Wwise 2026

C++, Blueprints, Wwise 2025.1, UE 5.8

- Designed a full Wwise audio architecture from scratch: RTPCs for player speed, occlusion, and reverb wetness; State groups for environment switching; Switch containers for surface-dependent footsteps.
- Built a four-bus mixing hierarchy (SFX, Ambient, Music, UI) with adaptive music transitions driven by game state.
- Integrated the middleware into UE5 and wired runtime parameter updates from gameplay code.

Spotify Data SQL Project 2025

Python, SQLite, SQL · github.com/saiweekend/Spotify-Data-SQL-Project

- Built an ETL pipeline ingesting music library data via the Spotify Web API into a normalised 3NF relational schema.
- Diagnosed a hard upstream blocker when Spotify deprecated audio-feature endpoints for new developer applications, and designed a deterministic local enrichment layer so the pipeline and downstream analysis remained fully reproducible.
- Wrote analytical SQL queries over the resulting dataset covering artist, genre, and audio-characteristic breakdowns.

Portfolio Site & In-Browser SQL Terminal 2025 – 2026

JavaScript, sql.js (SQLite/WASM), HTML/CSS · saihallie.com

- Designed and built a multi-page site from scratch with no framework, including a custom typography and animation system.
- Implemented a browser-based SQL terminal running SQLite compiled to WebAssembly, letting visitors execute live queries client-side against the project database.
- Deployed and maintained the production stack across GitHub Pages, Cloudflare CDN, and a custom domain, including cache invalidation and DNS configuration.

EDUCATION

Lesley University — Cambridge, Massachusetts, USA JULY 2023

B.S., Game Audio